

Lab 6

Intro to InSAR Time Series Methods

Why use time series methods?

- Single interferogram - Noisy displacement map
- Time series interferometry - deformation history, mitigates noise
- Phase “budget” - the factors that combine to produce measured phase

$$\phi_{ij}^{\text{obs}} = \phi_{ij}^{\text{def}} + \phi_{ij}^{\text{topo-res}} + \phi_{ij}^{\text{atm}} + \phi_{ij}^{\text{orbit}} + \phi_{ij}^{\text{noise}} + 2\pi k_{ij}$$

- k_{ij} here is integer ambiguity from phase unwrapping

Outputs: Average LOS velocity, displacement history $d(t)$, coherence/quality metrics, sometimes decomposed East/West and vertical components if using both ascending and descending interferograms.

Historical motivation: from InSAR snapshots to time series

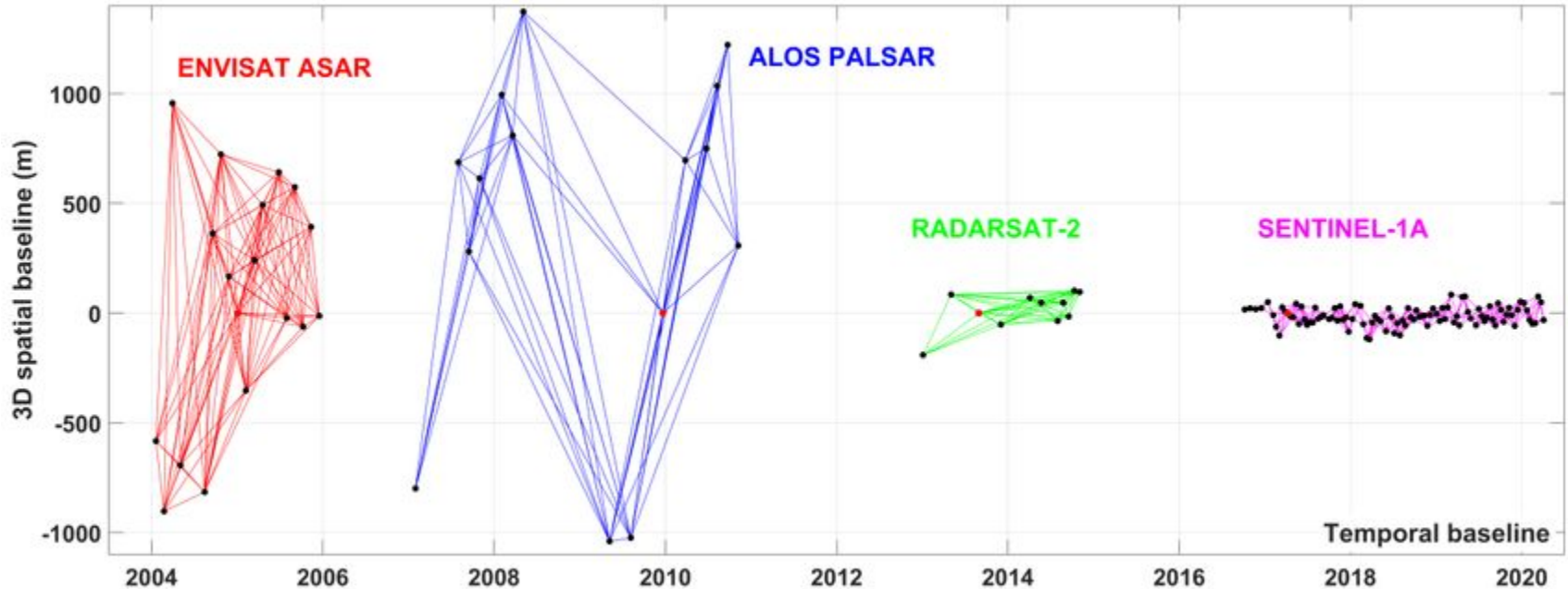
InSAR was spectacular for large events, but poor for subtle long-term deformation

Single interferogram has errors in decorrelation, atmosphere, orbit, and unwrapping

Two main early approaches (early 2000's): Persistent Scatterer and Small Baseline

- PS-InSAR (Ferretti et. al, 2001)
 - Exploit stable reflectors through long stacks
- SBAS (Berardino et. al, 2002)
 - Build many short-baseline interferograms to reduce spatial/temporal decorrelation
 - Invert interferogram network for deformation

A note on spatial baselines



- Modern Radar satellites have much tighter orbital controls
 - Anything over ~500m perpendicular baseline starts to significantly decorrelate, SBAS necessary. With modern constellations, data are already mostly small spatial baseline

Interferogram network as a graph

If we have acquisitions t_0, t_1, t_2, t_2 , and interferograms $(0,1),(1,2),(2,3),(0,3)$, unknown cumulative displacements are

$$\mathbf{m} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \quad \text{and we solve } \mathbf{d} = G\mathbf{m}$$
$$\begin{bmatrix} d_{01} \\ d_{12} \\ d_{23} \\ d_{03} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Each row of G is one interferogram. The -1 marks the earlier acquisition, the +1 the later acquisition, and the row computes displacement from. Phase-closure works because differences around a closed loop telescope to zero. We use t_0 as the reference point.

Interferogram network as a graph

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$$\begin{bmatrix} d_{01} \\ d_{12} \\ d_{23} \\ d_{03} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

- Each row of G encodes 1 interferogram
- Connected graph is required for coherent time series
- More edges provide redundancy
- Poor edges can be downweighted or discarded

Each row of G is one interferogram. The -1 marks the earlier acquisition, the +1 the later acquisition, and the row computes displacement from i to j . Phase-closure works because differences around a closed loop telescope to zero. We use t_0 as the reference point.

General least-squares inverse problem

$$\mathbf{d} = G\mathbf{m}$$

Time-series InSAR is an inverse problem with nuisance terms and regularization.

Weighted least-squares: $\hat{\mathbf{m}} = \arg \min_{\mathbf{m}} [(\mathbf{d} - G\mathbf{m})^T C_d^{-1} (\mathbf{d} - G\mathbf{m}) + \lambda^2 \|L\mathbf{m}\|^2]$

Linear algebra solution: $\hat{\mathbf{m}} = (G^T C_d^{-1} G + \lambda^2 L^T L)^{-1} G^T C_d^{-1} \mathbf{d}$

- $\mathbf{d}$: unwrapped phase
- G : design matrix of interferograms
- $\mathbf{m}$: displacement + noise
- C_d : data covariance (high coherence scenes can get higher weight)
- L : regularization (penalize improbable solutions)
- λ : strength of smoothing

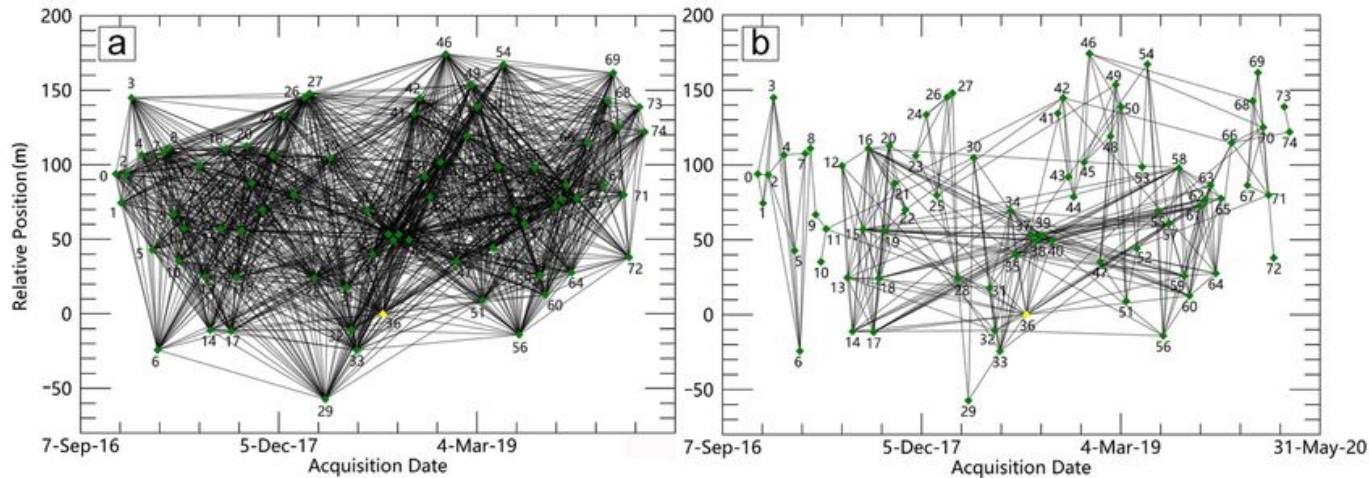
Main point: Time series is not simply read from the individual interferograms.

Time series is estimated from many pairwise differences under chosen assumptions

*Don't worry about fully understanding the math here, that's a topic for another course.

SBAS: Small Baseline Subset

- SBAS chooses interferograms that are likely to remain coherent
 - Small temporal baselines limit chances for surface changes to decorrelate scenes
 - Small perpendicular baselines limit chances for geometric differences to decorrelate scenes
 - Also reduce sensitivity to DEM errors
- Network may be sparse, but must remain connected for closure

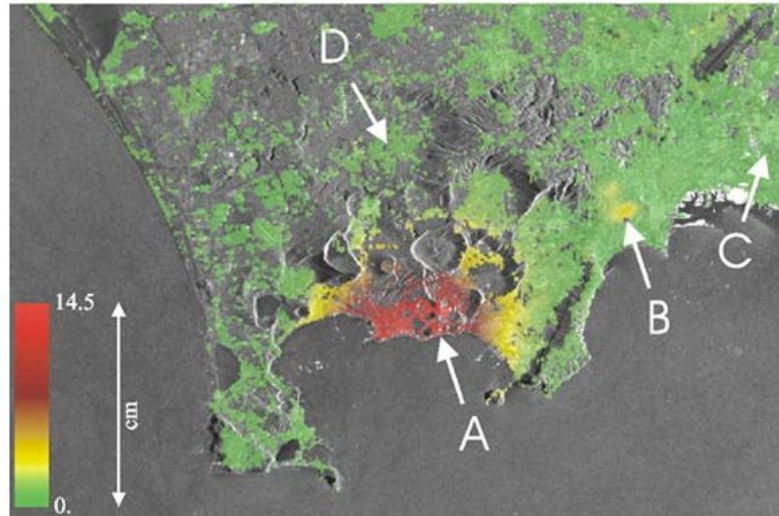


SBAS Workflow

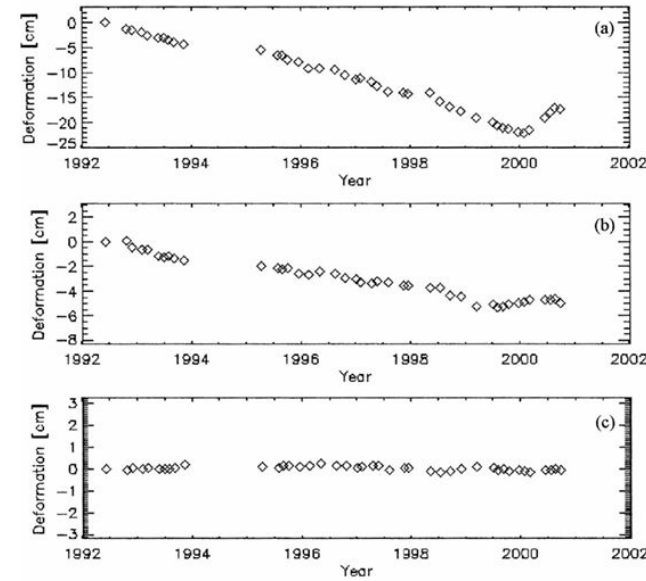
- Take SLC stack, coregister, choose small baseline pairs, form interferograms
- Filter phase, estimate coherence, and unwrap select interferograms
- Remove orbital ramps/atmosphere, invert network for $d(t)$, export

TABLE I
ERS-1/ERS-2 DATA USED. THE DIFFERENT
SB SUBSETS ARE HIGHLIGHTED

Mission	Orbit	Day	Month	Year	$B_{\perp}$	SB Subset
ERS1	4690	8	6	1992	0	1
ERS1	6694	26	10	1992	-124	1
ERS1	7195	30	11	1992	159	1
ERS1	8197	8	2	1993	-544	2
ERS1	8698	15	3	1993	-119	1
ERS1	9700	24	5	1993	-518	2
ERS1	10201	28	6	1993	-591	2
ERS1	10702	2	8	1993	-118	1
ERS1	11203	6	9	1993	31	1
ERS1	12205	15	11	1993	-58	1
ERS1	19563	12	4	1995	-525	2
ERS1	21066	26	7	1995	-149	1
ERS2	1393	27	7	1995	-185	1
ERS2	1894	31	8	1995	-555	2
ERS2	2395	5	10	1995	473	1
ERS2	3397	14	12	1995	19	1
ERS1	24072	21	2	1996	420	1
ERS1	25074	1	5	1996	400	1



(Berardino et. al, 2002)



PS-InSAR: Persistent Scatterers

PS-InSAR follows stable radar targets through time.

- A persistent scatterer is a pixel dominated by one stable scattering mechanism:
 - Building corner, metal object, bridge element, corner reflector
- Common candidate metrics are amplitude and phase stability
- PS is sparse but precise
 - Excellent for cities, infrastructure, rocky terrain
- PS processing estimates whether a point's residual phase behaves consistently after removing model terms.

PS and SBAS comparison

Model	Best targets	Pros	Cons
PS-InSAR	Stable strong scatterers	High precision	sparse
SBAS	Distributed coherent pixels	Better spatial coverage	Vulnerable to decorrelation and unwrapping errors
DS / hybrid	Statistically homogenous distributed scatterers	Improved spatial coverage over SBAS	More complex statistics and assumptions

Applications: Who actually uses this?

The value is not just pretty velocity maps; it is screening, prioritization, and monitoring

- Mining and post-mining deformation (active uplift/subsidence, abandoned mine collapse)
- Urban and infrastructure monitoring
- Hydrogeology (aquifer compaction, coastal subsidence)
- Landslides
- Volcanoes
- Tectonics
- Cryosphere/climate (permafrost thaw, glacial processes)

Who uses it?

Besides these academic studies, European Ground Motion Services is aimed at non-scientific end users.

<https://egms.land.copernicus.eu/>

Table 1. Classification of the EGMS applications published in the literature. In the References column, -B indicates the use of the EGMS Basic, -O the Ortho, and -C the Calibrated product. Product type is not specified in cases in which it is not mentioned by the authors.

Main topic	Sub-topic	References
Land subsidence	Coastal areas and relative sea level rise	Antoniadis et al. (2023) -O, Thiéblemont et al. (2023) -O, Melet et al. (2021) -O, Evelpidou et al. (2023) -O
	Groundwater exploitation	Hasan et al. (2023) -O, Mateos et al. (2024), Khan et al. (2024), Beccaro et al. (2023) -O, Ghaderpour et al. (2024) -C,O
Landslides	Local studies	Yishu (2022) -O, Nikolakopoulos et al. (2023) -O, Godone et al. (2023) -C, Dabiri and Nilfouroushan (2024) -B,C,O, Torre et al. (2024), Lau et al. (2024) -C,O
	Landslide inventories	Necula and Niculită (2023) -C, Necula and Niculită (2024) -C, Guinau et al. (2024) -C, Medici et al. (2024) -C
Geodynamics and tectonics	-	Atanasova and Nikolov (2023), Metzger et al. (2023) -O, Mildon et al. (2024) -O, Melichar et al. (2024) -O
Mining	Underground mining	Pawłuszek-Filipiak et al. (2023) -O, Ilieva et al. (2024)
	Open pit mining	Motagh et al. (2024), Tzampoglou and Loupasakis (2023) -O
	Abandoned mines	Yang et al. (2023) -O, Tinagli et al. (2024) -O

Example EGMS studies

Infrastructures	Bridges	Tonelli et al. (2023) -O, Eskandari and Scaioni (2023) -C,O
	Dams	Ruiz-Armenteros et al. (2023), Costantini et al. (2022) -B,C,O
	Civil infrastructure	Milenova Vasileva (2023) -O, Zêzere et al. (2024), Wang (2023) -O
	Gas storage	Fibbi et al. (2023) -C,O, Fibbi et al. (2024) -O, Rigamonti et al. (2024)
	Landfills	Negula et al. (2023) -O
Urban areas.	-	Mele et al. (2023) -C, Hlaváčová et al. (2023) -C,O, Shahbazi et al. (2022) -C, Shahbazi et al. (2024) -C, Rodríguez-Antuñano et al. (2023)
Buildings	-	
Archaeology and cultural heritage	-	Antonioli et al. (2023) -O, Agapiou et al. (2023) -O, Ioannidis et al. (2024) -O

Crosetto et. al, 2025